

Task 7 Report

Abaqus Wind Loading on Cooling Towers

Eneko ORHATEGARAY
Paul BROCVIELLE
Aiert CECCON
Achille LARREGLE

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Chapter 1

1. Objective

This Task 7 study expands the previous three-tower screening into a full structural comparison across the eight Task 6 scenarios and the three optimization algorithms SA, PSO, and BFGS. The engineering objective is to identify which Task 6 concepts remain convincing once the towers are checked under self-weight, one-direction wind loading, and linear buckling in Abaqus.

Chapter 2

2. Selection Rule and Scenario Matrix

The Task 7 matrix contains **24 refined presentation models**. For scenarios S1 to S7, the preferred choice is the lowest-area engineering-feasible Task 6 run for each scenario/algorithm pair. If no engineering-feasible run exists, the workflow falls back to the lowest-area mathematically compliant run. If neither exists, the best penalized run is kept only as a warning case. Scenario S8 is intentionally carried as warning-only because it has no compliant Task 6 solution in the current study.

Across the 24 selected cases, there are **16 engineering-feasible selections**, **3 mathematical fallbacks**, and **5 warning-only cases**.

Case	Status	Selection basis	Task 6 run	Decision mode	Task 6 area (m ²)
S1 / BFGS	Warning case	best_penalized_warning	6	radii	7782.01
S1 / PSO	Engineering-feasible	lowest_area_engineering_feasible	7	radii	7804.56
S1 / SA	Engineering-feasible	lowest_area_engineering_feasible	5	radii	7780.73
S2 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	2	radii	7741.67
S2 / PSO	Math fallback	lowest_area_mathematical_fallback	0	radii	9075.03
S2 / SA	Engineering-feasible	lowest_area_engineering_feasible	5	radii	7744.01
S3 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7741.69
S3 / PSO	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7743.34
S3 / SA	Engineering-feasible	lowest_area_engineering_feasible	6	radii	7742.77
S4 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	radii	7779.85
S4 / PSO	Warning case	best_penalized_warning	1	radii	7931.66
S4 / SA	Engineering-feasible	lowest_area_engineering_feasible	7	radii	7769.88
S5 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	0	heights	7742.56
S5 / PSO	Engineering-feasible	lowest_area_engineering_feasible	6	heights	7747.93
S5 / SA	Engineering-feasible	lowest_area_engineering_feasible	6	heights	7749.52
S6 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	heights	7741.72
S6 / PSO	Engineering-feasible	lowest_area_engineering_feasible	3	heights	7748.25
S6 / SA	Engineering-feasible	lowest_area_engineering_feasible	0	heights	7783.24
S7 / BFGS	Engineering-feasible	lowest_area_engineering_feasible	1	joint	7740.66
S7 / PSO	Math fallback	lowest_area_mathematical_fallback	9	joint	8324.25
S7 / SA	Math fallback	lowest_area_mathematical_fallback	9	joint	7890.85
S8 / BFGS	Warning case	best_penalized_warning	5	joint	7813.24
S8 / PSO	Warning case	best_penalized_warning	8	joint	7899.07
S8 / SA	Warning case	best_penalized_warning	0	joint	7718.27

Task 7 candidate cooling-tower profiles selected from Task 6

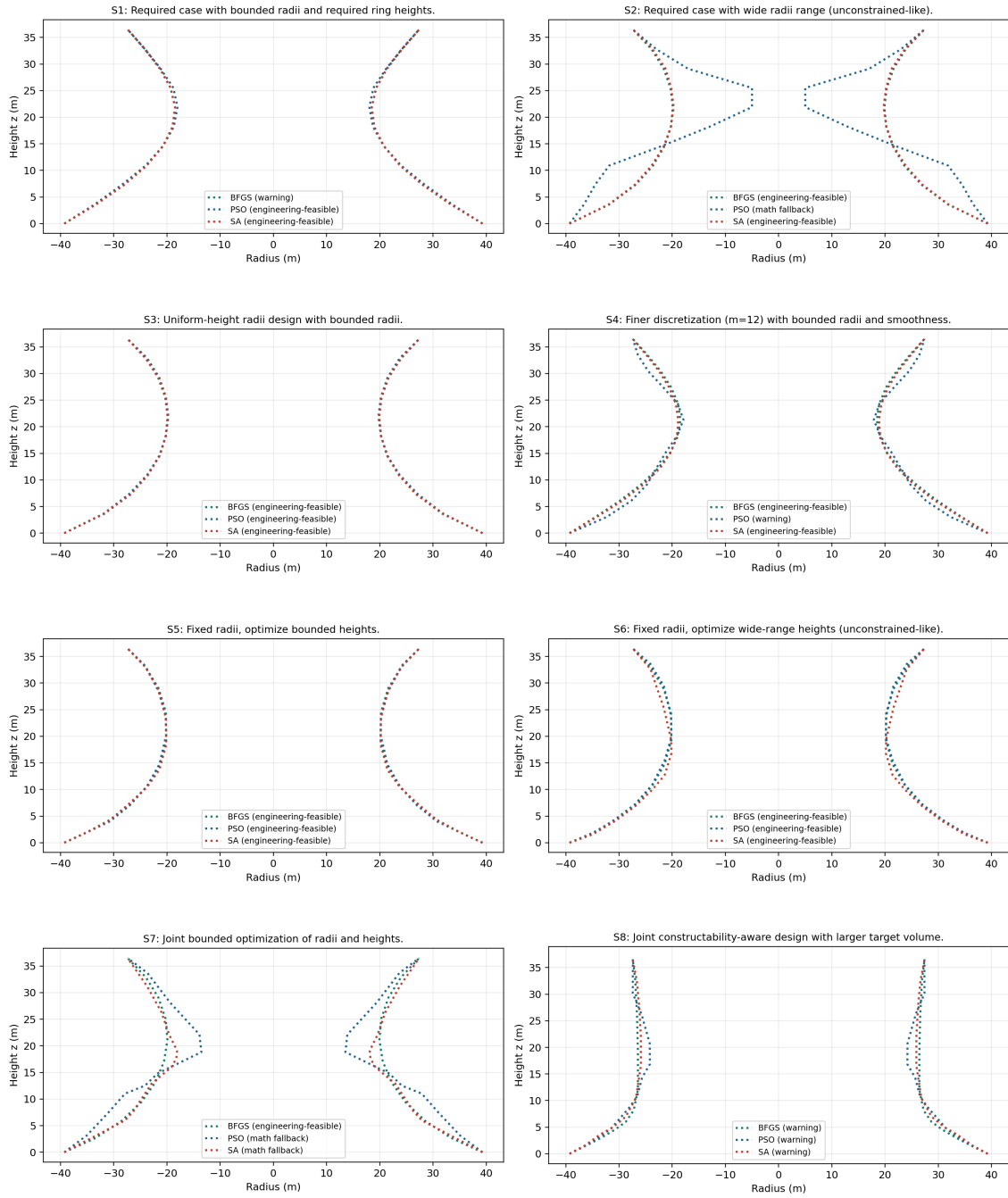


Figure 2.1: Task 7 candidate profiles

Chapter 3

3. Material, Loading, and CAE Setup

Each model is built from the selected Task 6 meridian profile and revolved into a thin-shell cooling-tower surface; Task 7 keeps the exact Task 6 geometry scale.

The global coordinate convention is Y-up, with X/Z as the horizontal plane.

Material assumptions are uniform across all cases: equivalent reinforced concrete with $E = 33$ GPa, $\nu = 0.20$, density $\rho = 2500 \text{ kg/m}^3$, and shell thickness $t = 0.20 \text{ m}$.

Loading combines self-weight and one-direction wind. The reference wind speed is 30.0 m/s , giving $q = 562.5 \text{ Pa}$ from $q = 0.5 \rho_{air} V^2$.

Wind is modeled as external pressure on the shell with windward pressure and leeward suction (not internal cabin pressure). The circumferential pressure law is:

$$C_p(\theta) = \text{clip}(0.8 \cos \theta, -0.5, 0.8), \quad p(\theta) = q C_p(\theta)$$

Boundary conditions represent the tower support at the foundation/piler ring: base nodes are fixed ($U_x = U_y = U_z = 0$), while the shell remains free elsewhere.

Each model uses one static step (STATIC_WIND) for combined loading and one linear buckling step (BUCKLING) to extract the first instability modes about the preloaded state.

Chapter 4

4. Refined Structural Comparison Across All 24 Cases

Case	Status	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
S1 / BFGS	Warning case	7782.01	4.264	2.262	20.907
S1 / PSO	Engineering-feasible	7804.56	4.126	2.244	21.300
S1 / SA	Engineering-feasible	7780.73	4.627	2.368	21.481
S2 / BFGS	Engineering-feasible	7741.67	10.202	3.867	23.563
S2 / PSO	Math fallback	9075.03	94.040	10.268	-3.229
S2 / SA	Engineering-feasible	7744.01	9.968	3.818	24.549
S3 / BFGS	Engineering-feasible	7741.69	10.153	3.834	23.529
S3 / PSO	Engineering-feasible	7743.34	9.850	3.794	23.496
S3 / SA	Engineering-feasible	7742.77	9.036	3.526	23.834
S4 / BFGS	Engineering-feasible	7779.85	3.831	2.282	20.267
S4 / PSO	Warning case	7931.66	12.073	4.541	22.491
S4 / SA	Engineering-feasible	7769.88	4.287	2.296	19.753
S5 / BFGS	Engineering-feasible	7742.56	8.695	4.148	22.725
S5 / PSO	Engineering-feasible	7747.93	11.481	4.941	19.566
S5 / SA	Engineering-feasible	7749.52	10.229	4.642	23.232
S6 / BFGS	Engineering-feasible	7741.72	10.522	5.312	24.772
S6 / PSO	Engineering-feasible	7748.25	11.907	6.705	21.611
S6 / SA	Engineering-feasible	7783.24	14.270	7.690	19.128
S7 / BFGS	Engineering-feasible	7740.66	10.105	4.134	24.948
S7 / PSO	Math fallback	8324.25	28.467	9.258	23.712
S7 / SA	Math fallback	7890.85	12.541	6.711	26.587
S8 / BFGS	Warning case	7813.24	10.319	5.446	19.273
S8 / PSO	Warning case	7899.07	9.235	4.493	27.335
S8 / SA	Warning case	7718.27	8.163	4.955	23.837

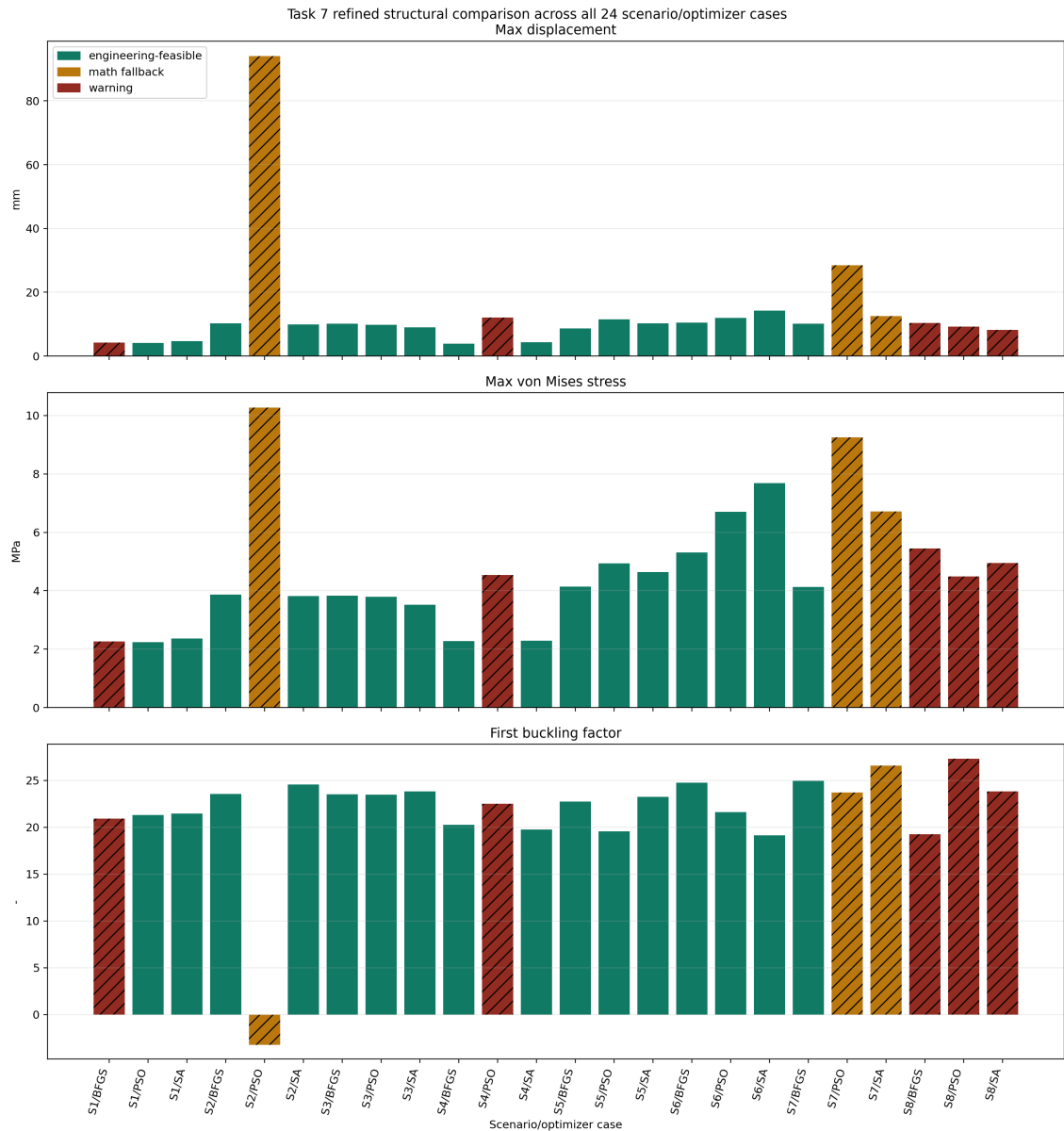


Figure 4.1: Task 7 refined comparison metrics

Figure 4.1 stacks displacement, stress, and buckling vertically to improve visibility and avoid compressed labels.

The first reading pass should prioritize buckling spread, then check whether displacement and stress trends confirm the same structural leader.

Chapter 5

5. Methodological Clarity and Global Ranking

5.1 5.1 Methodological Clarity

The global ranking is a rank-based weighted score on the refined results, designed to stay robust when one or two cases have extreme values.

For each metric, rank 1 is best and rank 24 is worst, with deterministic tie-breaks by scenario/algorithm identifiers.

$$r_i^{(k)} = \text{rank}_k(i)$$

$$S_i = 0.45 r_i^{(b)} + 0.25 r_i^{(d)} + 0.20 r_i^{(s)} + 0.10 r_i^{(a)}$$

$$J_i = S_i + P_i, \quad P_i \in \{0, 10, 20\}$$

Here $r_i^{(b)}$ is the buckling rank (higher buckling is better), $r_i^{(d)}$ is the displacement rank, $r_i^{(s)}$ is the stress rank, and $r_i^{(a)}$ is the area rank (lower is better for the last three).

Penalty points are 0 for engineering-feasible, 10 for mathematical fallback, and 20 for warning cases. The best model is the one with the lowest J_i .

5.2 5.2 Global Weighted Top-5

Rank	Case	Status	Weighted score	Penalty	Buckling factor	Max disp. (mm)	Max stress (MPa)	Area (m ²)
1	S7 / BFGS	Engineering-feasible	6.750	0.0	24.948	10.105	4.134	7740.66
2	S3 / SA	Engineering-feasible	7.050	0.0	23.834	9.036	3.526	7742.77
3	S2 / SA	Engineering-feasible	7.500	0.0	24.549	9.968	3.818	7744.01
4	S3 / PSO	Engineering-feasible	9.650	0.0	23.496	9.850	3.794	7743.34
5	S2 / BFGS	Engineering-feasible	9.850	0.0	23.563	10.202	3.867	7741.67

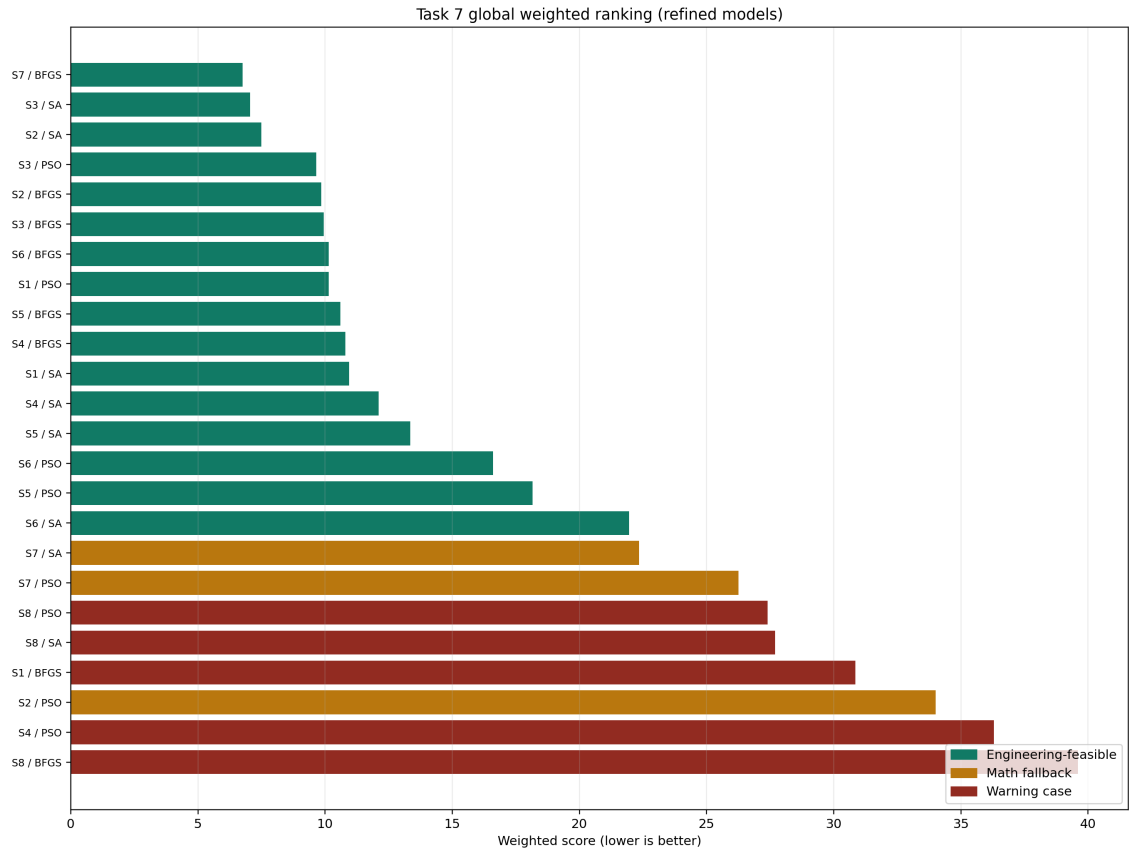


Figure 5.1: Task 7 weighted ranking

5.3 5.3 Top-3 by Criterion

5.3.1 5.3.1 Raw Top-3 (all statuses)

This raw criterion table keeps all statuses visible, so warning and fallback cases can appear if they are numerically extreme on a single indicator.

Criterion	Rank	Case	Status	Value
Buckling factor	1	S8 / PSO	Warning case	27.335
Buckling factor	2	S7 / SA	Math fallback	26.587
Buckling factor	3	S7 / BFGS	Engineering-feasible	24.948
Max displacement	1	S4 / BFGS	Engineering-feasible	3.831 mm
Max displacement	2	S1 / PSO	Engineering-feasible	4.126 mm
Max displacement	3	S1 / BFGS	Warning case	4.264 mm
Max stress	1	S1 / PSO	Engineering-feasible	2.244 MPa
Max stress	2	S1 / BFGS	Warning case	2.262 MPa
Max stress	3	S4 / BFGS	Engineering-feasible	2.282 MPa
Task 6 area	1	S8 / SA	Warning case	7718.27 m ²
Task 6 area	2	S7 / BFGS	Engineering-feasible	7740.66 m ²
Task 6 area	3	S2 / BFGS	Engineering-feasible	7741.67 m ²

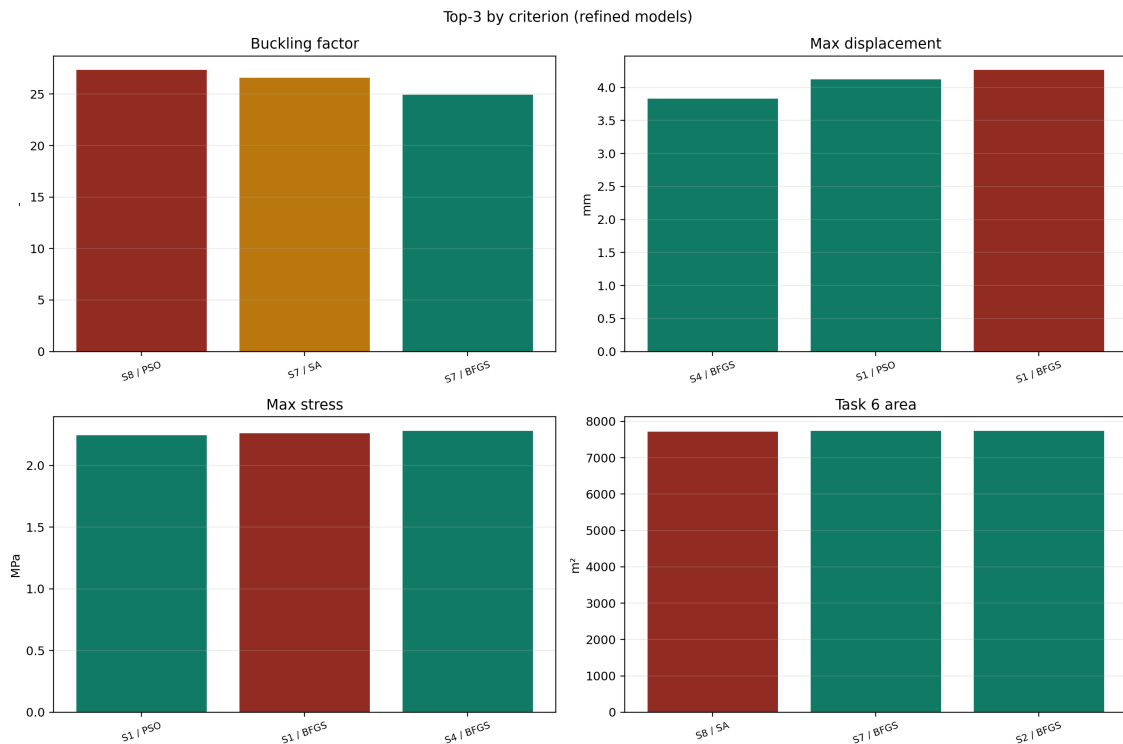


Figure 5.2: Task 7 criterion leaders

5.3.2 5.3.2 Engineering-eligible Top-3

This decision-grade table filters to engineering-feasible entries only.

Criterion	Rank	Case	Status	Value
Buckling factor	1	S7 / BFGS	Engineering-feasible	24.948
Buckling factor	2	S6 / BFGS	Engineering-feasible	24.772
Buckling factor	3	S2 / SA	Engineering-feasible	24.549
Max displacement	1	S4 / BFGS	Engineering-feasible	3.831 mm
Max displacement	2	S1 / PSO	Engineering-feasible	4.126 mm
Max displacement	3	S4 / SA	Engineering-feasible	4.287 mm
Max stress	1	S1 / PSO	Engineering-feasible	2.244 MPa
Max stress	2	S4 / BFGS	Engineering-feasible	2.282 MPa
Max stress	3	S4 / SA	Engineering-feasible	2.296 MPa
Task 6 area	1	S7 / BFGS	Engineering-feasible	7740.66 m ²
Task 6 area	2	S2 / BFGS	Engineering-feasible	7741.67 m ²
Task 6 area	3	S3 / BFGS	Engineering-feasible	7741.69 m ²

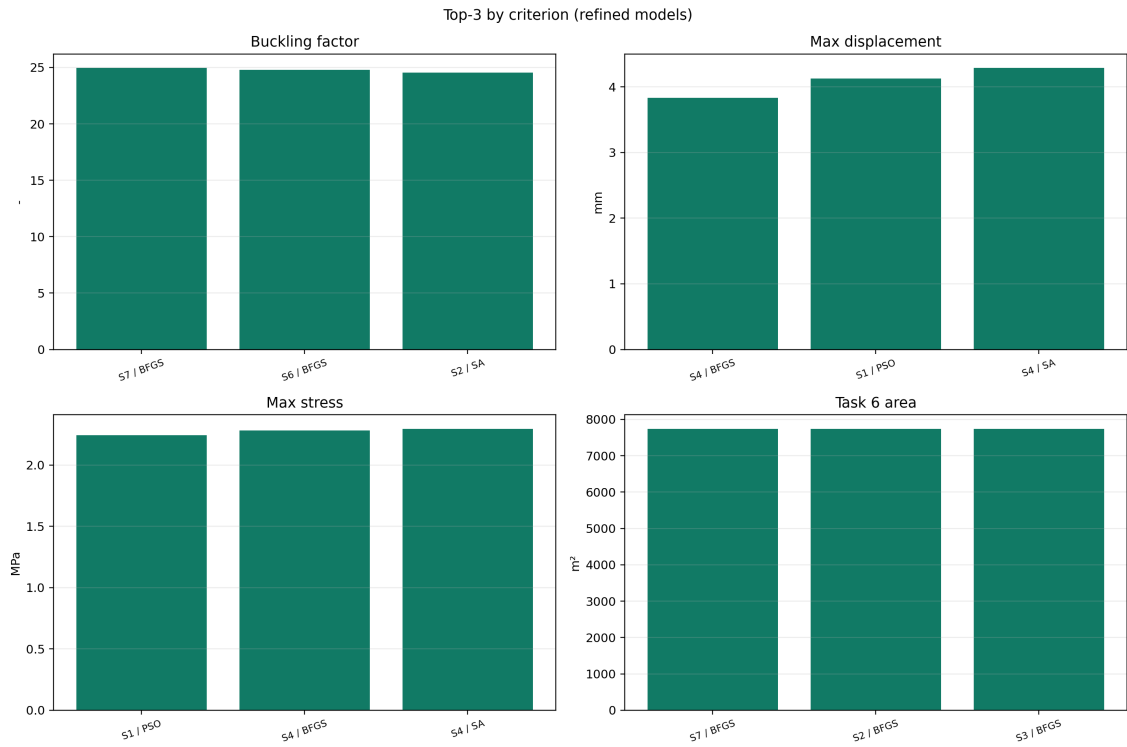


Figure 5.3: Task 7 engineering criterion leaders

Chapter 6

6. Scenario-by-Scenario Comparison

6.1 S1

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Warning case	30.850	7782.01	4.264	2.262	20.907
PSO	Engineering-feasible	10.150	7804.56	4.126	2.244	21.300
SA	Engineering-feasible	10.950	7780.73	4.627	2.368	21.481

Winner: S1 / PSO with weighted score 10.150; gap to second place (S1 / SA) is 0.800.

Spread/Risk: displacement spans 4.126 to 4.627 mm, stress spans 2.244 to 2.368 MPa, and buckling spans 20.907 to 21.481.

Status caveat: warning-only entries are present for BFGS and are not decision-grade designs.

Load dominance: winner ratios are disp wind/gravity 0.107 and stress wind/gravity 0.073; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

6.2 S2

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	9.850	7741.67	10.202	3.867	23.563
PSO	Math fallback	34.000	9075.03	94.040	10.268	-3.229
SA	Engineering-feasible	7.500	7744.01	9.968	3.818	24.549

Winner: S2 / SA with weighted score 7.500; gap to second place (S2 / BFGS) is 2.350.

Spread/Risk: displacement spans 9.968 to 94.040 mm, stress spans 3.818 to 10.268 MPa, and buckling spans -3.229 to 24.549.

Status caveat: mathematical fallback entries are present for PSO.

Load dominance: winner ratios are disp wind/gravity 0.028 and stress wind/gravity 0.042; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

6.3 S3

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	9.950	7741.69	10.153	3.834	23.529
PSO	Engineering-feasible	9.650	7743.34	9.850	3.794	23.496
SA	Engineering-feasible	7.050	7742.77	9.036	3.526	23.834

Winner: S3 / SA with weighted score 7.050; gap to second place (S3 / PSO) is 2.600.

Spread/Risk: displacement spans 9.036 to 10.153 mm, stress spans 3.526 to 3.834 MPa, and buckling spans 23.496 to 23.834.

Status caveat: all three entries are engineering-feasible.

Load dominance: winner ratios are disp wind/gravity 0.034 and stress wind/gravity 0.049; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: winner ranking is not yet convergence-locked on this scenario; keep this as screening guidance until mesh convergence is improved.

6.4 S4

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.800	7779.85	3.831	2.282	20.267
PSO	Warning case	36.300	7931.66	12.073	4.541	22.491
SA	Engineering-feasible	12.100	7769.88	4.287	2.296	19.753

Winner: S4 / BFGS with weighted score 10.800; gap to second place (S4 / SA) is 1.300.

Spread/Risk: displacement spans 3.831 to 12.073 mm, stress spans 2.282 to 4.541 MPa, and buckling spans 19.753 to 22.491.

Status caveat: warning-only entries are present for PSO and are not decision-grade designs.

Load dominance: winner ratios are disp wind/gravity 0.078 and stress wind/gravity 0.079; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

6.5 S5

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.600	7742.56	8.695	4.148	22.725
PSO	Engineering-feasible	18.150	7747.93	11.481	4.941	19.566
SA	Engineering-feasible	13.350	7749.52	10.229	4.642	23.232

Winner: S5 / BFGS with weighted score 10.600; gap to second place (S5 / SA) is 2.750.

Spread/Risk: displacement spans 8.695 to 11.481 mm, stress spans 4.148 to 4.941 MPa, and buckling spans 19.566 to 23.232.

Status caveat: all three entries are engineering-feasible.

Load dominance: winner ratios are disp wind/gravity 0.034 and stress wind/gravity 0.044; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: winner ranking is not yet convergence-locked on this scenario; keep this as screening guidance until mesh convergence is improved.

6.6 S6

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	10.150	7741.72	10.522	5.312	24.772
PSO	Engineering-feasible	16.600	7748.25	11.907	6.705	21.611
SA	Engineering-feasible	21.950	7783.24	14.270	7.690	19.128

Winner: S6 / BFGS with weighted score 10.150; gap to second place (S6 / PSO) is 6.450.

Spread/Risk: displacement spans 10.522 to 14.270 mm, stress spans 5.312 to 7.690 MPa, and buckling spans 19.128 to 24.772.

Status caveat: all three entries are engineering-feasible.

Load dominance: winner ratios are disp wind/gravity 0.026 and stress wind/gravity 0.030; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: winner ranking is not yet convergence-locked on this scenario; keep this as screening guidance until mesh convergence is improved.

6.7 S7

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Engineering-feasible	6.750	7740.66	10.105	4.134	24.948
PSO	Math fallback	26.250	8324.25	28.467	9.258	23.712
SA	Math fallback	22.350	7890.85	12.541	6.711	26.587

Winner: S7 / BFGS with weighted score 6.750; gap to second place (S7 / SA) is 15.600.

Spread/Risk: displacement spans 10.105 to 28.467 mm, stress spans 4.134 to 9.258 MPa, and buckling spans 23.712 to 26.587.

Status caveat: mathematical fallback entries are present for PSO, SA.

Load dominance: winner ratios are disp wind/gravity 0.029 and stress wind/gravity 0.036; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: scenario fairness is reduced by mixed compliance statuses; ranking penalties keep the comparison visible but do not make warning/fallback cases equivalent to engineering-feasible designs.

6.8 S8

Algorithm	Status	Weighted score	Task 6 area (m ²)	Max disp. (mm)	Max stress (MPa)	Buckling factor
BFGS	Warning case	39.600	7813.24	10.319	5.446	19.273
PSO	Warning case	27.400	7899.07	9.235	4.493	27.335
SA	Warning case	27.700	7718.27	8.163	4.955	23.837

Winner: S8 / PSO with weighted score 27.400; gap to second place (S8 / SA) is 0.300.

Spread/Risk: displacement spans 8.163 to 10.319 mm, stress spans 4.493 to 5.446 MPa, and buckling spans 19.273 to 27.335.

Status caveat: warning-only entries are present for BFGS, PSO, SA and are not decision-grade designs.

Load dominance: winner ratios are disp wind/gravity 0.069 and stress wind/gravity 0.041; dominant drivers are displacement **gravity** and stress **gravity**.

Critical note: winner ranking is not yet convergence-locked on this scenario; keep this as screening guidance until mesh convergence is improved.

Chapter 7

7. Convergence Summary

The verification workflow solves both coarse and refined meshes for every case. The acceptance criteria are: displacement change below 5.0%, stress change below 5.0%, and first buckling-factor change below 10.0%.

At the current stage, **0 of 24 comparison pairs** satisfy all three convergence criteria.

Because the study is currently at 0/24 all-pass convergence, the ranking should be interpreted as comparative screening quality, not final structural qualification.

Case	Status	Delta disp. (%)	Pass	Delta stress (%)	Pass	Delta buckling (%)	Pass	All pass
S1 / BFGS	Warning case	8.66	no	3.95	yes	-51.38	no	no
S1 / PSO	Engineering-feasible	17.03	no	8.32	no	-51.39	no	no
S1 / SA	Engineering-feasible	9.59	no	2.24	yes	-49.67	no	no
S2 / BFGS	Engineering-feasible	27.17	no	0.60	yes	-48.05	no	no
S2 / PSO	Math fallback	-11.47	no	-60.11	no	-5.64	yes	no
S2 / SA	Engineering-feasible	25.49	no	-1.19	yes	-46.75	no	no
S3 / BFGS	Engineering-feasible	26.57	no	-0.73	yes	-48.06	no	no
S3 / PSO	Engineering-feasible	27.31	no	1.44	yes	-50.01	no	no
S3 / SA	Engineering-feasible	21.16	no	-3.57	yes	-46.37	no	no
S4 / BFGS	Engineering-feasible	-5.76	no	7.01	no	-51.16	no	no
S4 / PSO	Warning case	16.53	no	-3.15	yes	-158.30	no	no
S4 / SA	Engineering-feasible	-7.69	no	1.43	yes	-49.02	no	no
S5 / BFGS	Engineering-feasible	-5.89	no	10.45	no	-46.31	no	no
S5 / PSO	Engineering-feasible	1.17	yes	23.43	no	-49.25	no	no
S5 / SA	Engineering-feasible	-2.76	yes	5.00	no	-45.85	no	no
S6 / BFGS	Engineering-feasible	2.08	yes	20.49	no	-41.81	no	no
S6 / PSO	Engineering-feasible	17.42	no	39.74	no	-54.85	no	no
S6 / SA	Engineering-feasible	15.36	no	38.65	no	-55.49	no	no
S7 / BFGS	Engineering-feasible	17.31	no	0.69	yes	-43.58	no	no
S7 / PSO	Math fallback	11.87	no	-19.73	no	-240.70	no	no
S7 / SA	Math fallback	-6.17	no	-19.68	no	-63.19	no	no
S8 / BFGS	Warning case	2.71	yes	10.91	no	-53.42	no	no
S8 / PSO	Warning case	7.91	no	-0.58	yes	-49.86	no	no
S8 / SA	Warning case	5.04	no	15.88	no	-53.02	no	no

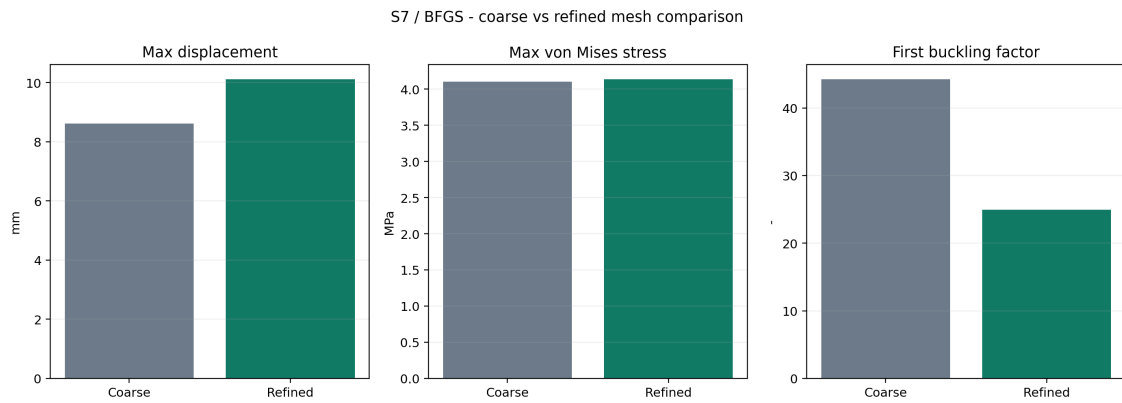


Figure 7.1: Winner mesh comparison

The global winner is therefore reported as **provisional**: the ranking is valid for comparative screening, but convergence evidence is not yet strong enough for a final structural sign-off.

Chapter 8

8. Wind-vs-Self-Weight Decomposition (Refined Cases)

To separate load influence explicitly, each refined comparison tower is re-solved with gravity-only and wind-only static steps.

The combined-load ranking remains unchanged; decomposition is used as an interpretation layer.

Across all 24 cases, displacement is wind-dominant in 1 cases and stress is wind-dominant in 0 cases.

Median wind-to-gravity ratios are 0.048 for displacement and 0.043 for stress, with IQR ranges 0.028-0.083 (displacement) and 0.040-0.072 (stress).

Tail sensitivity remains important: P90 wind-to-gravity ratios are 0.179 for displacement and 0.081 for stress.

Critical interpretation: most towers remain gravity-led in global reaction, while wind can still control local displacement/stress behavior for specific geometries. This is why the report keeps both decomposition evidence and weighted ranking instead of collapsing to one indicator.

Because convergence is currently 0/24 all-pass, decomposition trends are used as high-value screening evidence but not as final qualification proof.

Case	Status	Disp gravity (mm)	Disp wind (mm)	Wind/Gravity disp ratio	Disp driver	Stress gravity (MPa)	Stress wind (MPa)	Wind/Gravity stress ratio	Stress driver
S1 / BFGS	Warning case	4.050	0.299	0.074	gravity	2.224	0.166	0.074	gravity
S1 / PSO	Engineering-feasible	3.919	0.418	0.107	gravity	2.211	0.161	0.073	gravity
S1 / SA	Engineering-feasible	4.411	0.303	0.069	gravity	2.320	0.165	0.071	gravity
S2 / BFGS	Engineering-feasible	10.020	0.278	0.028	gravity	3.818	0.161	0.042	gravity
S2 / PSO	Math fallback	44.473	55.530	1.249	wind	9.986	2.478	0.248	gravity
S2 / SA	Engineering-feasible	9.790	0.277	0.028	gravity	3.769	0.160	0.042	gravity
S3 / BFGS	Engineering-feasible	9.970	0.277	0.028	gravity	3.784	0.162	0.043	gravity
S3 / PSO	Engineering-feasible	9.675	0.271	0.028	gravity	3.745	0.157	0.042	gravity
S3 / SA	Engineering-feasible	8.836	0.301	0.034	gravity	3.473	0.171	0.049	gravity
S4 / BFGS	Engineering-feasible	3.621	0.282	0.078	gravity	2.238	0.176	0.079	gravity
S4 / PSO	Warning case	11.848	1.946	0.164	gravity	4.472	0.185	0.041	gravity
S4 / SA	Engineering-feasible	4.074	0.303	0.074	gravity	2.251	0.185	0.082	gravity
S5 / BFGS	Engineering-feasible	8.507	0.288	0.034	gravity	4.093	0.181	0.044	gravity
S5 / PSO	Engineering-feasible	11.380	0.288	0.025	gravity	4.932	0.209	0.042	gravity
S5 / SA	Engineering-feasible	10.116	0.257	0.025	gravity	4.591	0.180	0.039	gravity
S6 / BFGS	Engineering-feasible	10.353	0.274	0.026	gravity	5.267	0.159	0.030	gravity
S6 / PSO	Engineering-feasible	11.752	0.270	0.023	gravity	6.672	0.141	0.021	gravity
S6 / SA	Engineering-feasible	14.102	0.285	0.020	gravity	7.657	0.133	0.017	gravity
S7 / BFGS	Engineering-feasible	9.928	0.284	0.029	gravity	4.104	0.149	0.036	gravity
S7 / PSO	Math fallback	23.968	5.240	0.219	gravity	8.941	0.987	0.110	gravity
S7 / SA	Math fallback	11.478	2.124	0.185	gravity	6.564	0.352	0.054	gravity
S8 / BFGS	Warning case	10.116	1.007	0.100	gravity	5.367	0.195	0.036	gravity
S8 / PSO	Warning case	9.130	0.629	0.069	gravity	4.458	0.181	0.041	gravity
S8 / SA	Warning case	8.019	0.489	0.061	gravity	4.888	0.228	0.047	gravity

Chapter 9

9. Warning Cases from Scenario S8

Scenario S8 is kept intentionally as a warning family. None of the Task 6 S8 runs are mathematically compliant or engineering-feasible, so these Abaqus models are useful only as structural cautionary examples and not as candidate towers for recommendation.

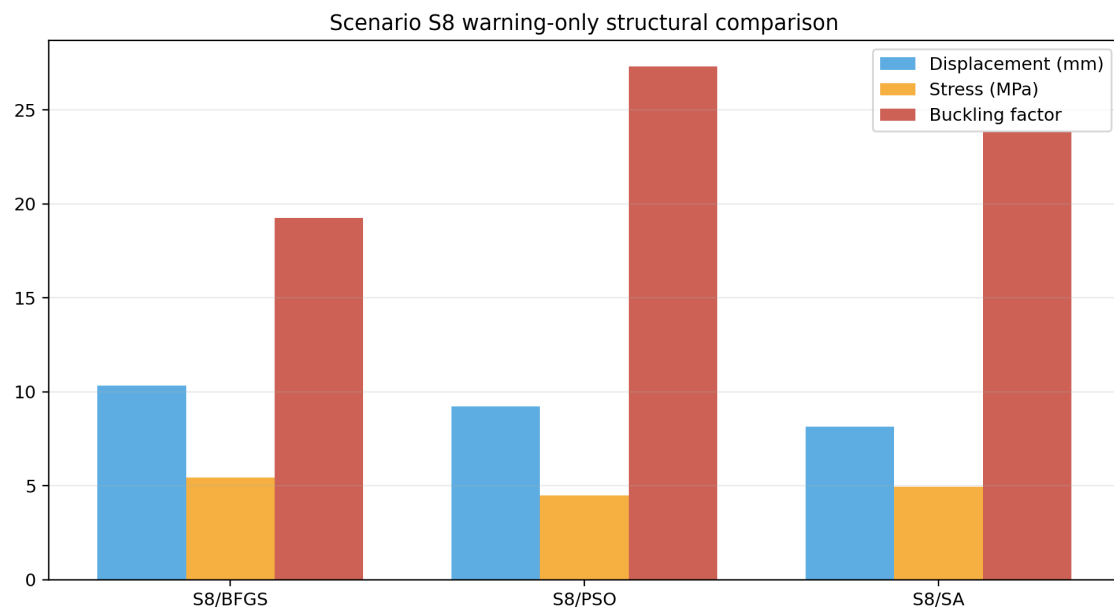


Figure 9.1: Scenario S8 warning metrics

Chapter 10

10. Field Visualizations of the Global Top-5

All detailed field figures are rendered from actual Abaqus ODB data with Python, not from Abaqus screenshots.

Simulation geometry remains true-scale from Task 6. For readability, the rendered figures use a display-only vertical exaggeration of 1.35x and a consistent left-to-right wind-view policy. Wind arrows are bound to the configured physical wind axis (+X), so they indicate actual load direction rather than decorative annotation.

Per-case camera/arrow verification is enforced numerically through `plot_view_audit.csv`: the camera is perpendicular to wind direction, wind projects left-to-right on screen, and arrows stay outside the tower silhouette.

10.1 S7 / BFGS

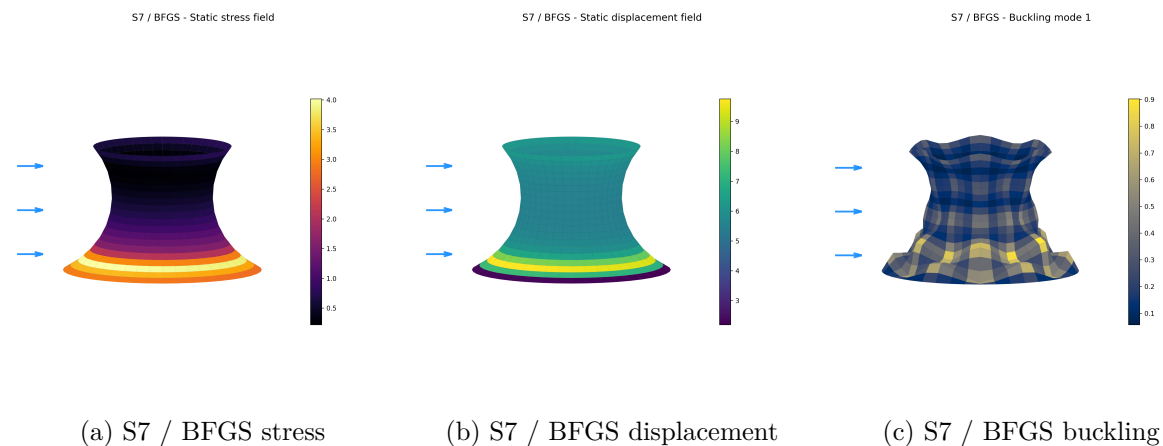


Figure 10.1: Combined visual fields

Performance Discussion (Rank 1): The S7 / BFGS model (Engineering-feasible) achieved an overall weighted score of 6.750. It demonstrates strong structural integrity with a first buckling factor of 24.948. Under the applied wind load, the maximum observed displacement is bounded at 10.105 mm, and peak von Mises stresses reach 4.134 MPa. As the top-ranked

candidate, this geometry offers the most convincing balance of minimal material footprint, acceptable deflection, and a high margin against linear buckling.

10.2 S3 / SA

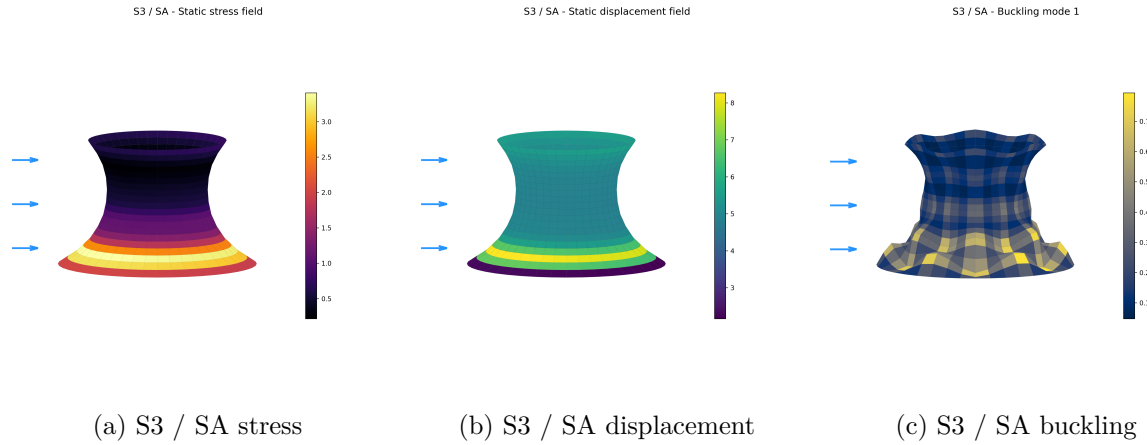


Figure 10.2: Combined visual fields

Performance Discussion (Rank 2): The S3 / SA model (Engineering-feasible) achieved an overall weighted score of 7.050. It demonstrates strong structural integrity with a first buckling factor of 23.834. Under the applied wind load, the maximum observed displacement is bounded at 9.036 mm, and peak von Mises stresses reach 3.526 MPa. As a high-ranking runner-up, it presents an extremely competitive alternative, trading minimal surface area differences for robust stress and displacement behavior.

10.3 S2 / SA

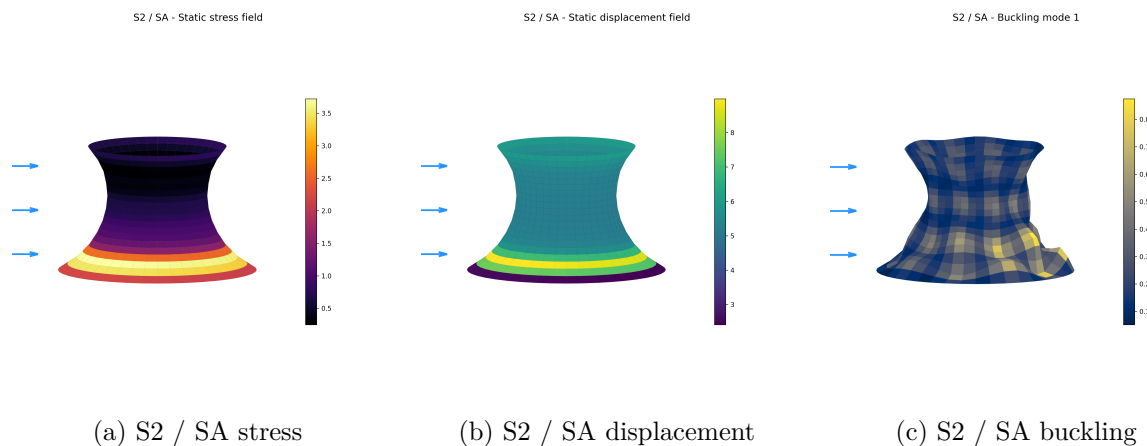


Figure 10.3: Combined visual fields

Performance Discussion (Rank 3): The S2 / SA model (Engineering-feasible) achieved an overall weighted score of 7.500. It demonstrates strong structural integrity with a first buckling

factor of 24.549. Under the applied wind load, the maximum observed displacement is bounded at 9.968 mm, and peak von Mises stresses reach 3.818 MPa. As a high-ranking runner-up, it presents an extremely competitive alternative, trading minimal surface area differences for robust stress and displacement behavior.

10.4 S3 / PSO

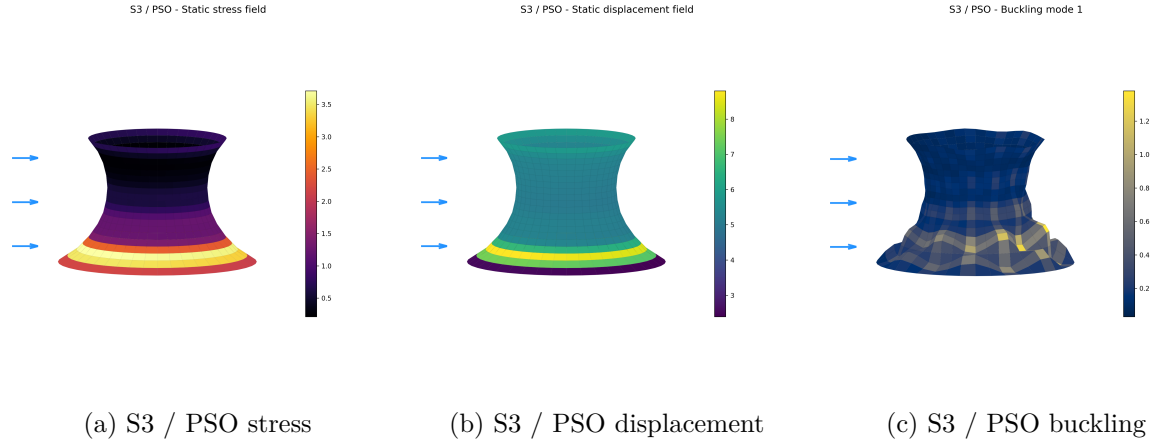


Figure 10.4: Combined visual fields

Performance Discussion (Rank 4): The S3 / PSO model (Engineering-feasible) achieved an overall weighted score of 9.650. It demonstrates strong structural integrity with a first buckling factor of 23.496. Under the applied wind load, the maximum observed displacement is bounded at 9.850 mm, and peak von Mises stresses reach 3.794 MPa. Although ranking slightly lower within the top 5, it remains a structurally sound and convincing concept that safely satisfies engineering constraints.

10.5 S2 / BFGS

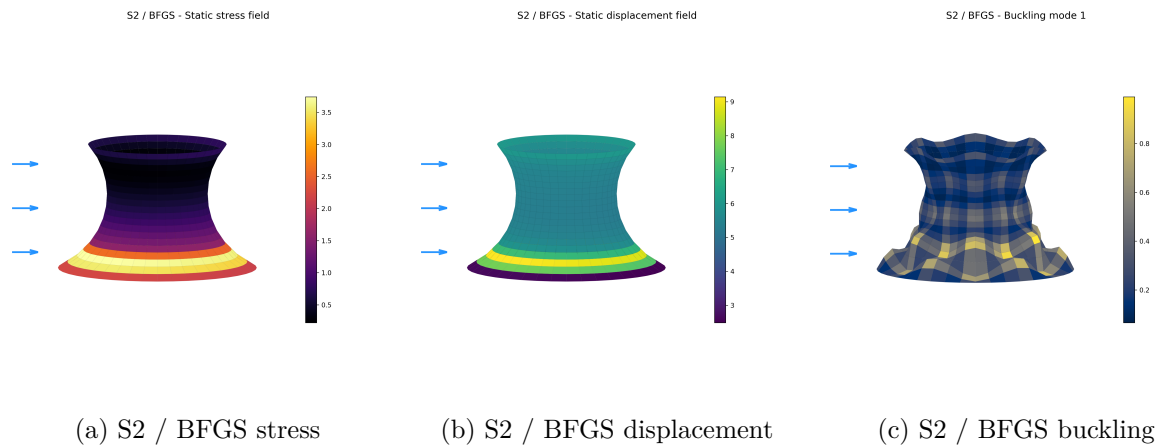


Figure 10.5: Combined visual fields

Performance Discussion (Rank 5): The S2 / BFGS model (Engineering-feasible) achieved an overall weighted score of 9.850. It demonstrates strong structural integrity with a first buckling factor of 23.563. Under the applied wind load, the maximum observed displacement is bounded at 10.202 mm, and peak von Mises stresses reach 3.867 MPa. Although ranking slightly lower within the top 5, it remains a structurally sound and convincing concept that safely satisfies engineering constraints.

Chapter 11

11. Annex: Complete Field Visualizations

This section contains the field visualizations for the remaining 19 candidates out of the total 24 refined presentation models, organized by Scenario.

11.1 Scenario S1

11.1.1 S1 / BFGS

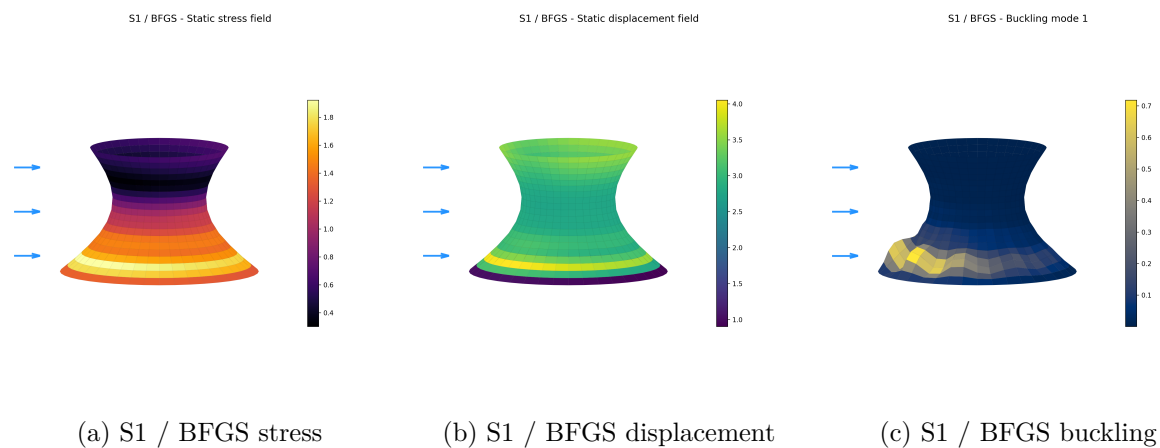


Figure 11.1: Combined visual fields

11.1.2 S1 / PSO

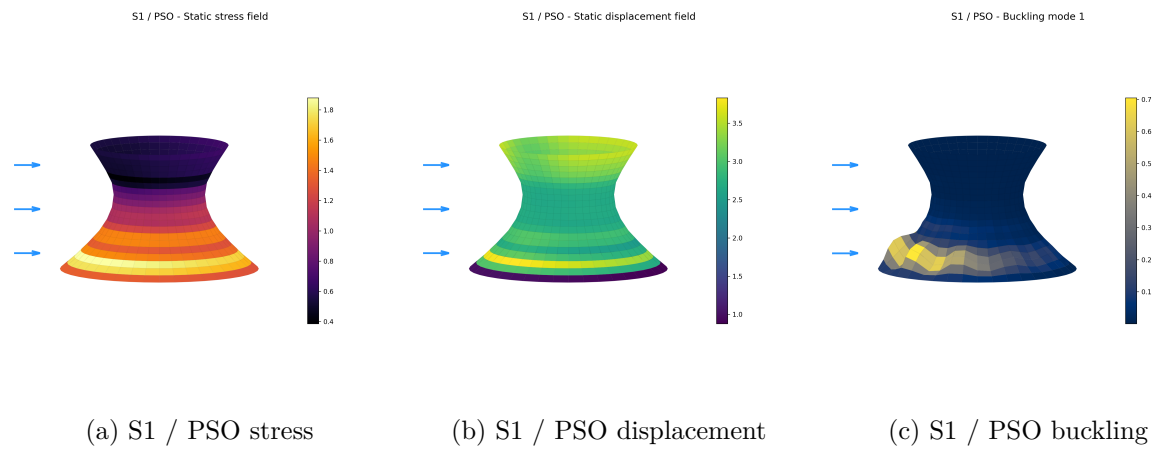


Figure 11.2: Combined visual fields

11.1.3 S1 / SA

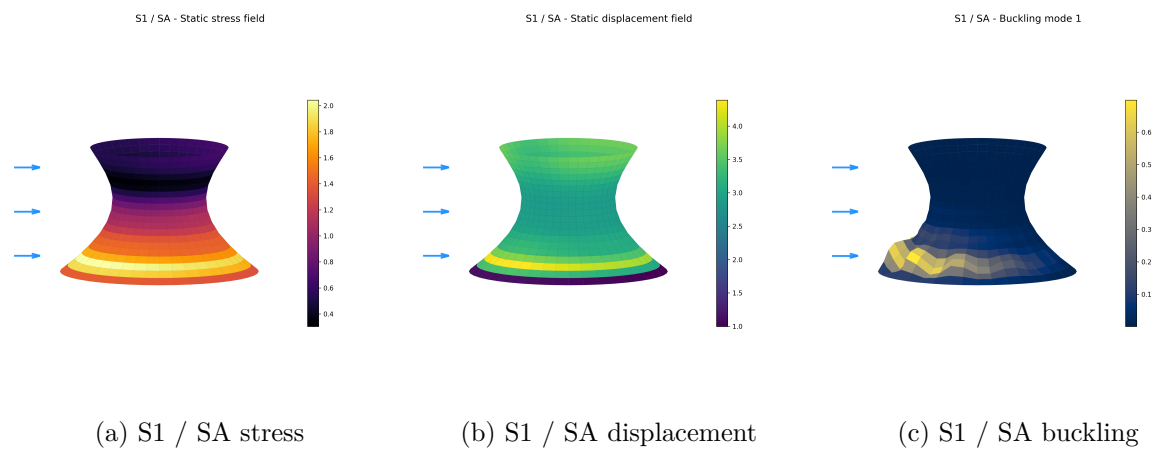


Figure 11.3: Combined visual fields

11.2 Scenario S2

11.2.1 S2 / PSO

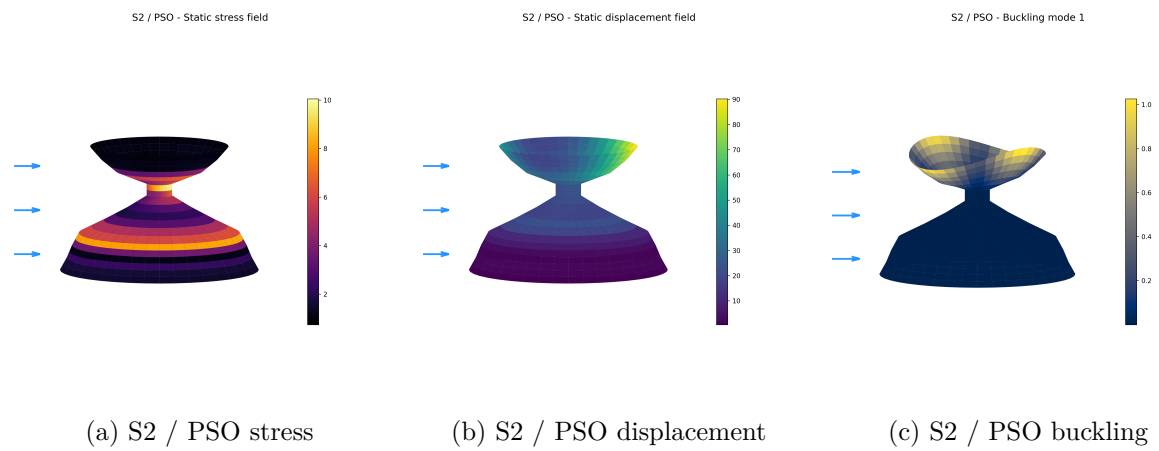


Figure 11.4: Combined visual fields

11.3 Scenario S3

11.3.1 S3 / BFGS

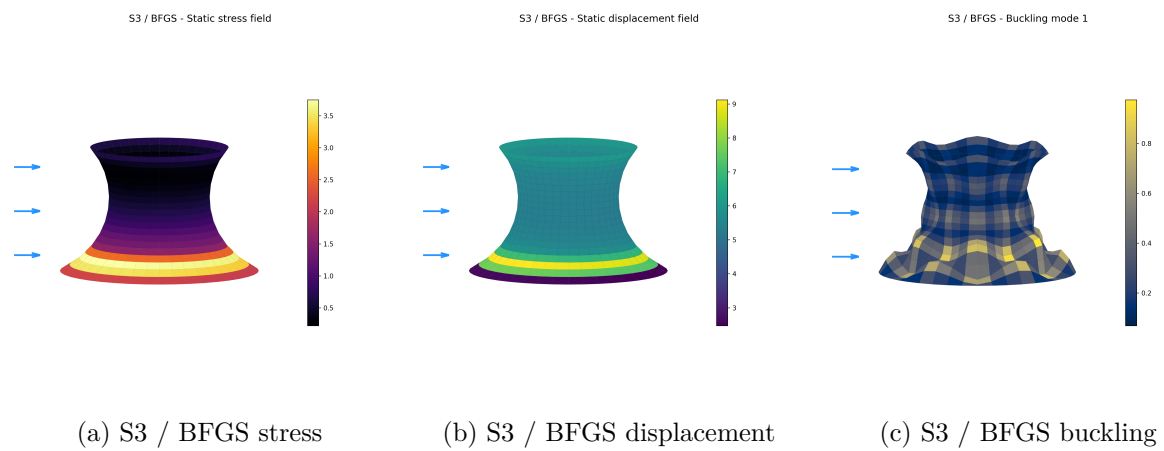


Figure 11.5: Combined visual fields

11.4 Scenario S4

11.4.1 S4 / BFGS

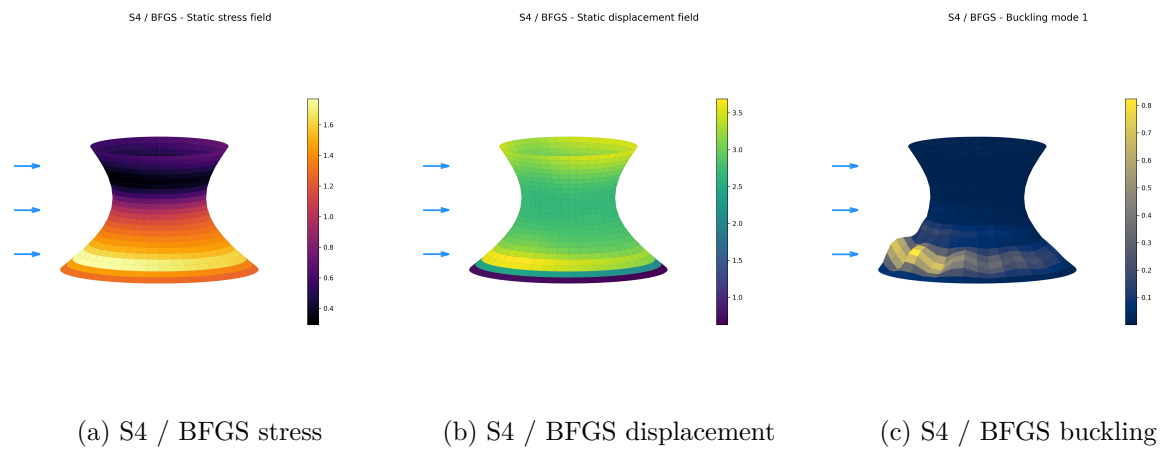


Figure 11.6: Combined visual fields

11.4.2 S4 / PSO

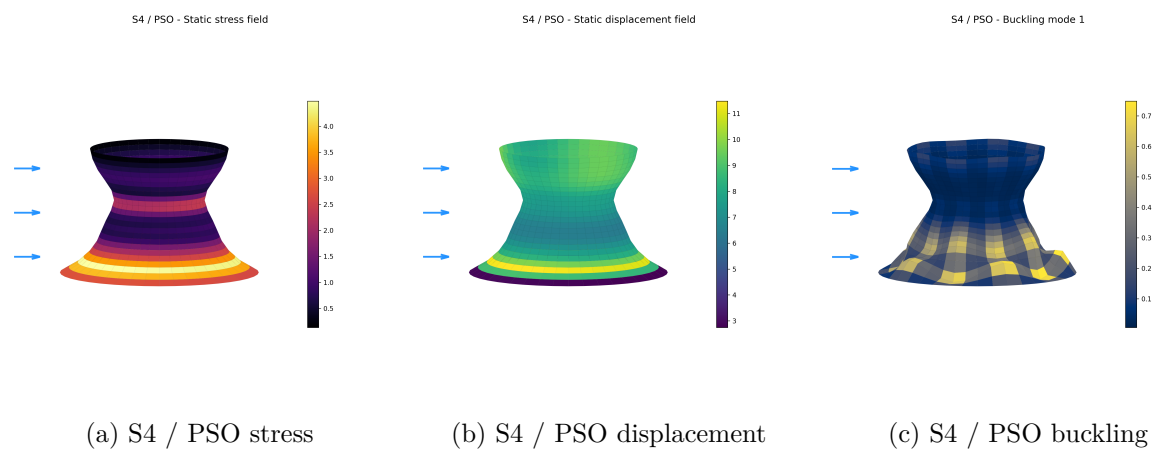


Figure 11.7: Combined visual fields

11.4.3 S4 / SA

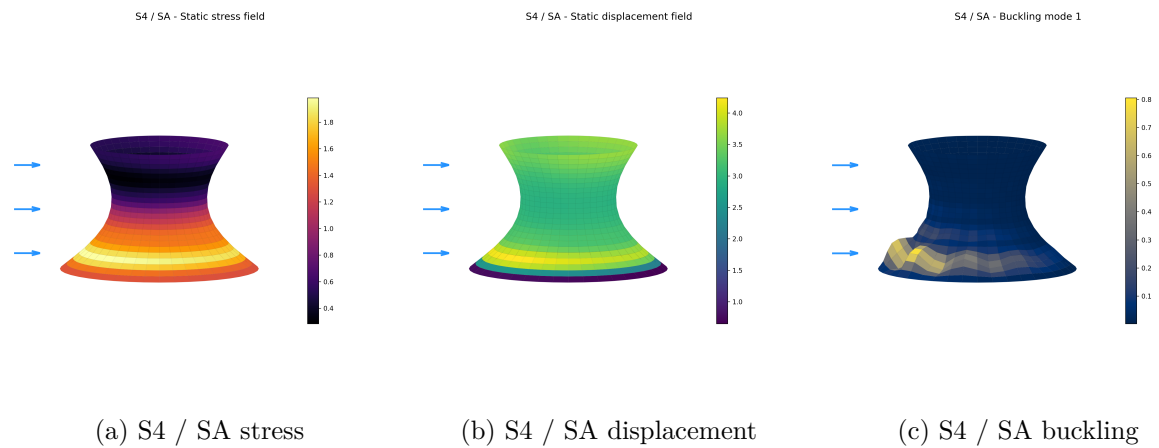


Figure 11.8: Combined visual fields

11.5 Scenario S5

11.5.1 S5 / BFGS

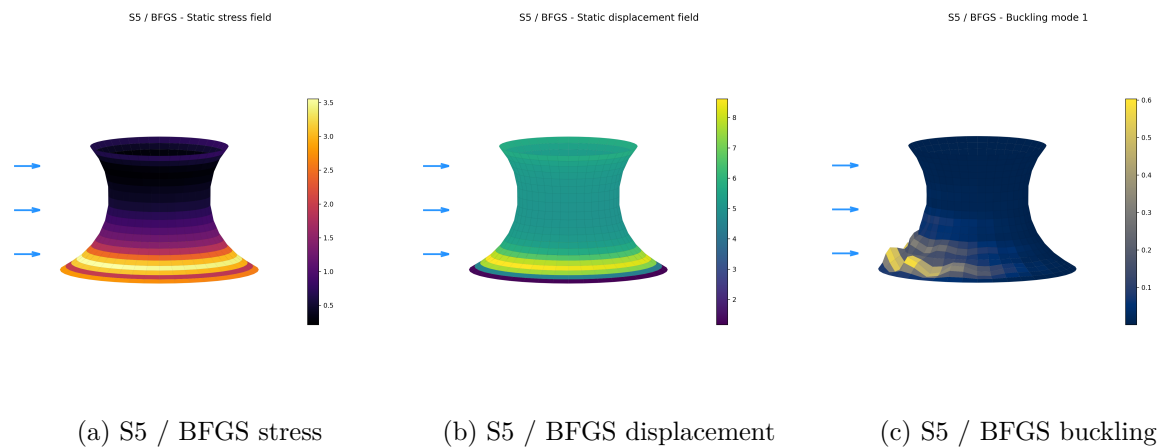


Figure 11.9: Combined visual fields

11.5.2 S5 / PSO

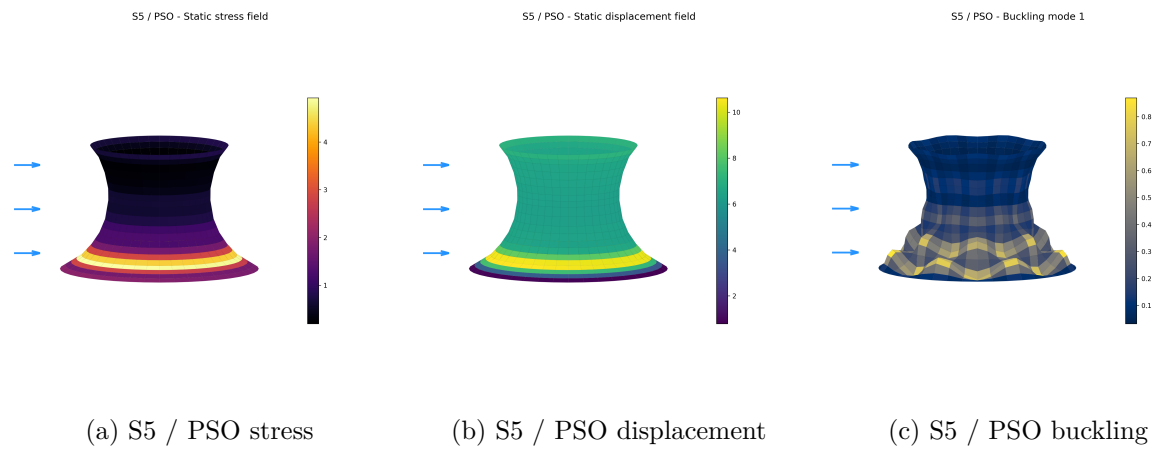


Figure 11.10: Combined visual fields

11.5.3 S5 / SA

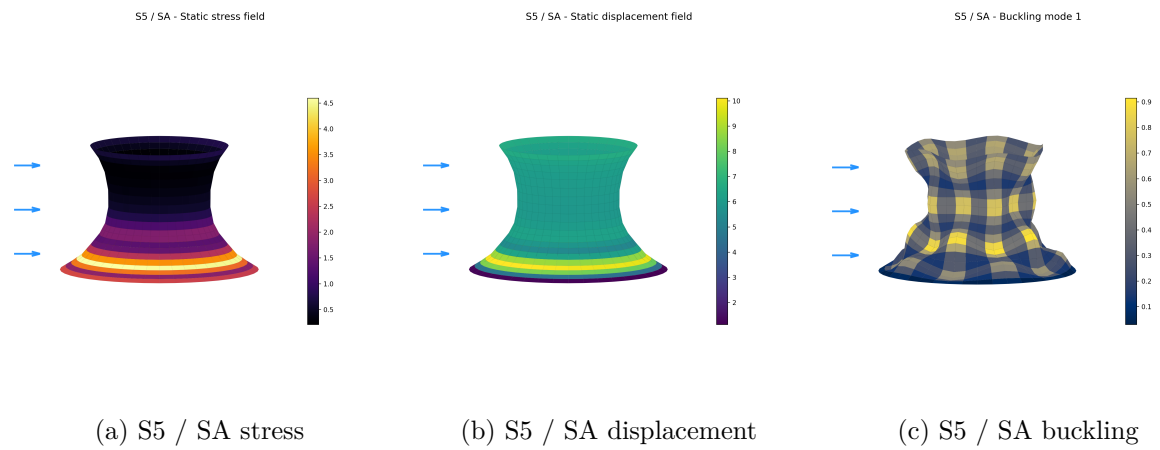


Figure 11.11: Combined visual fields

11.6 Scenario S6

11.6.1 S6 / BFGS

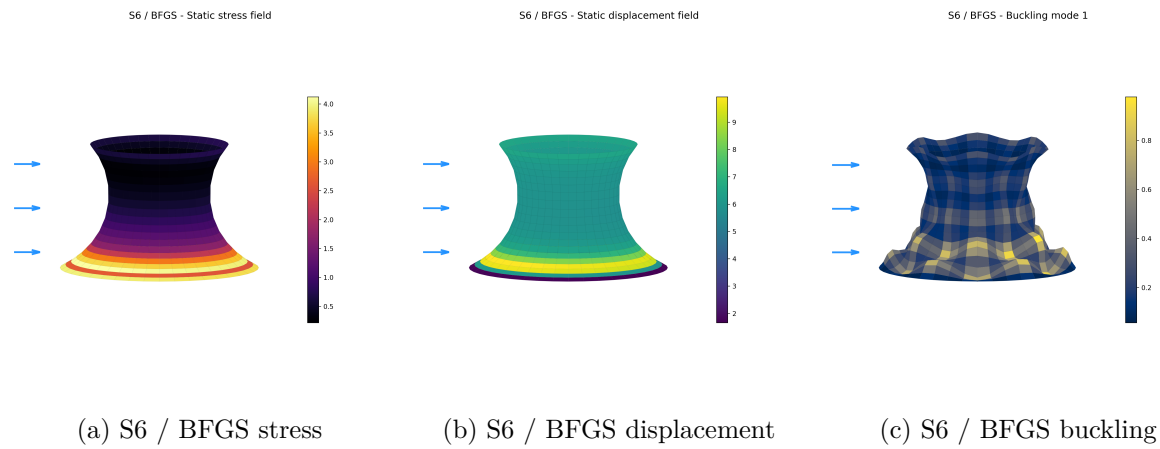


Figure 11.12: Combined visual fields

11.6.2 S6 / PSO

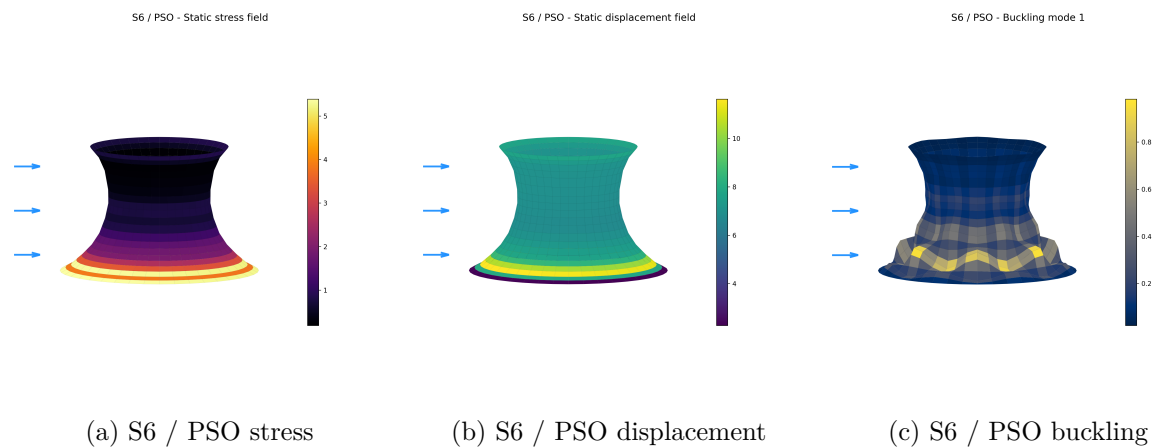


Figure 11.13: Combined visual fields

11.6.3 S6 / SA

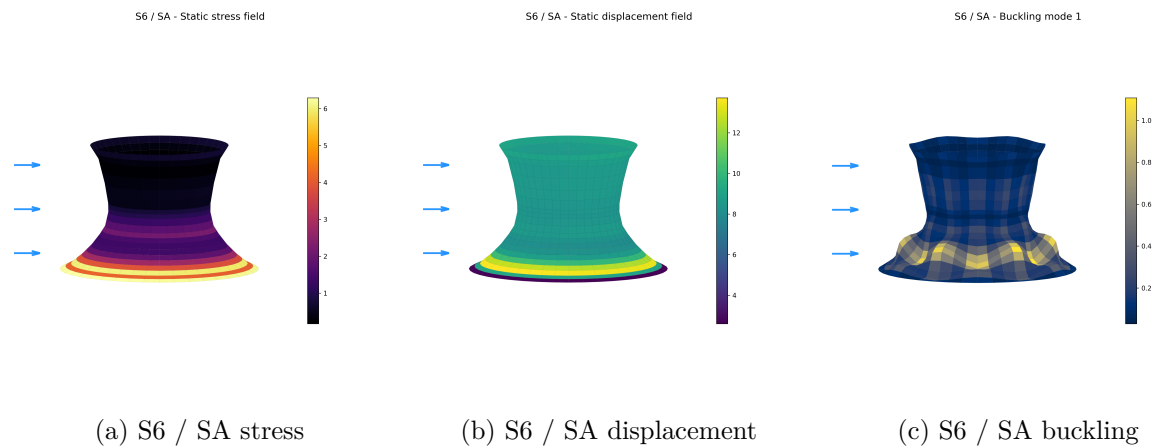


Figure 11.14: Combined visual fields

11.7 Scenario S7

11.7.1 S7 / PSO

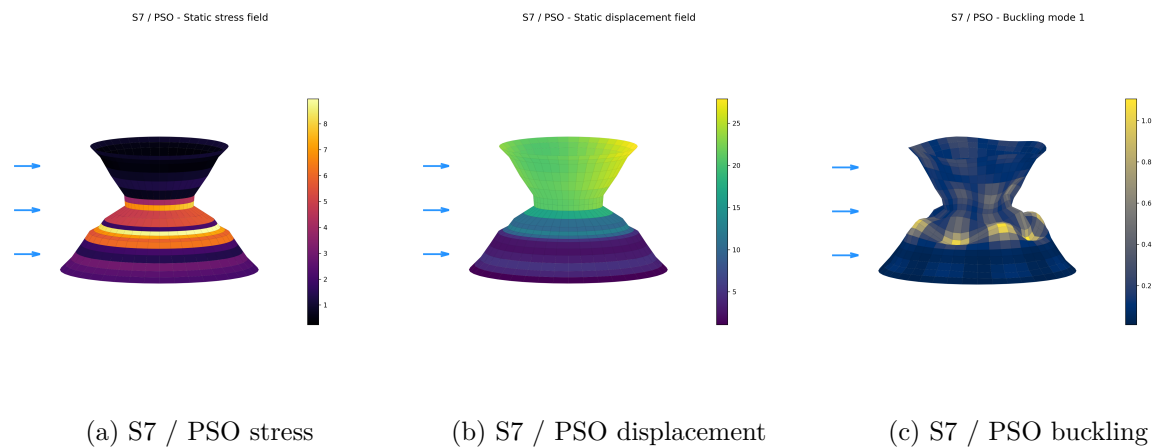


Figure 11.15: Combined visual fields

11.7.2 S7 / SA

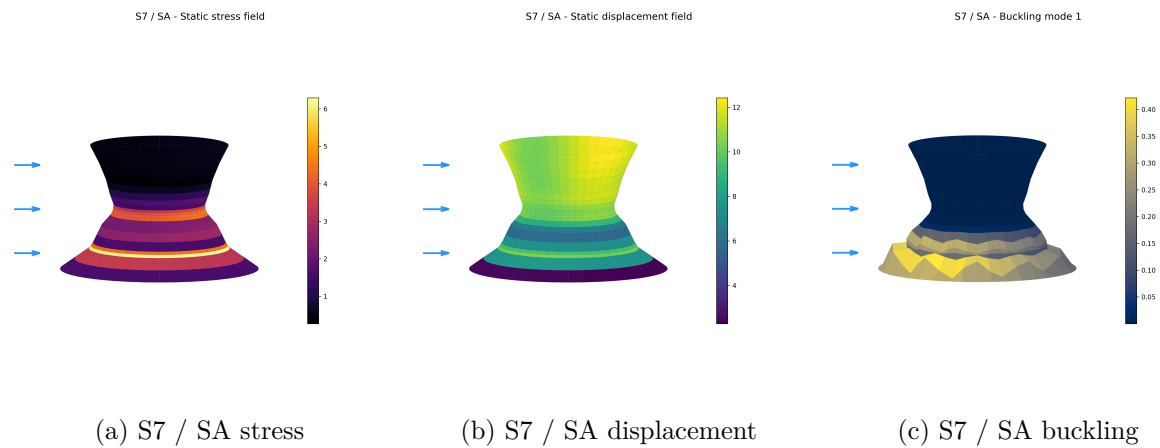


Figure 11.16: Combined visual fields

11.8 Scenario S8

11.8.1 S8 / BFGS

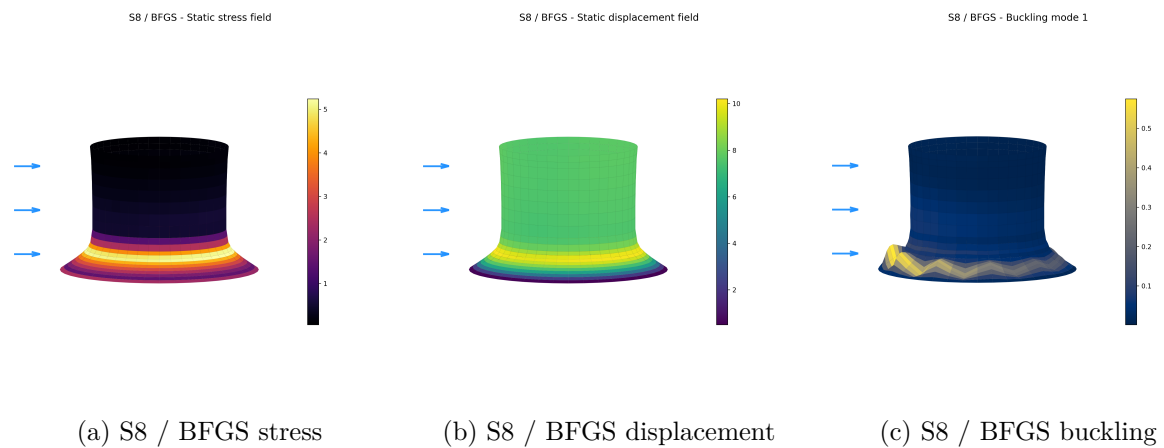


Figure 11.17: Combined visual fields

11.8.2 S8 / PSO

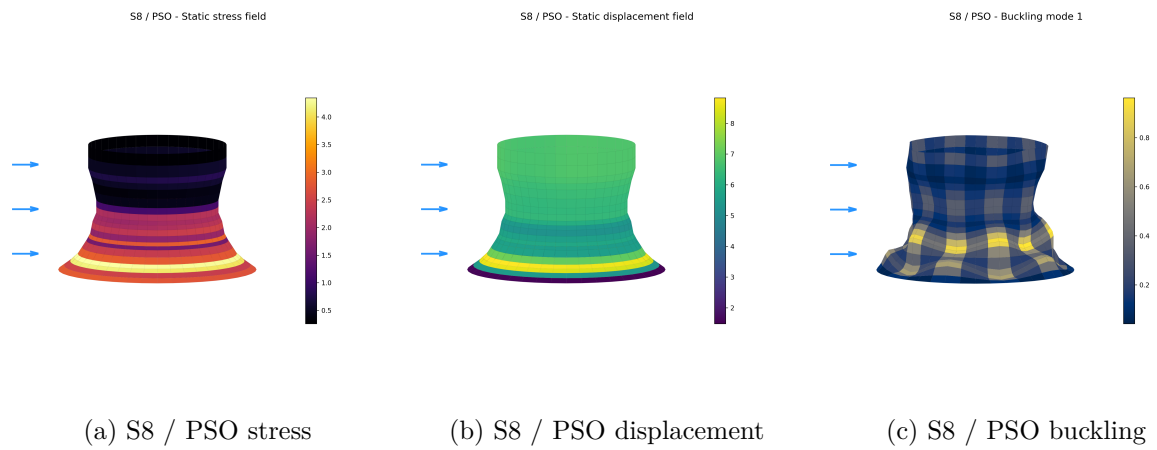


Figure 11.18: Combined visual fields

11.8.3 S8 / SA

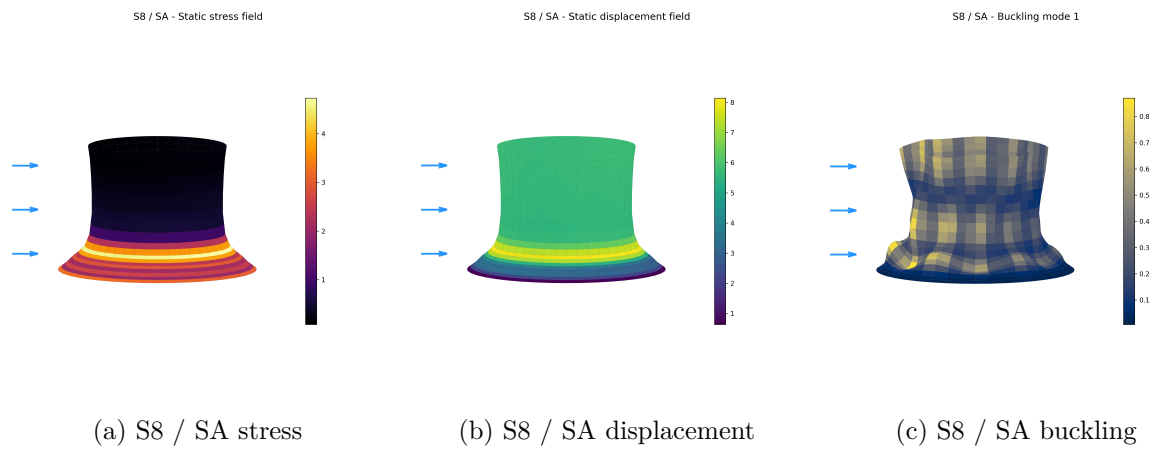


Figure 11.19: Combined visual fields

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- [1] CEN, *Eurocode 2: Design of concrete structures – part 1-1: General rules and rules for buildings*, 2004.
- [2] CEN, *Eurocode 1: Actions on structures – part 1-4: Wind actions*, 2005.
- [3] S. P. Timoshenko and S. Woinowsky-Krieger, *Theory of Plates and Shells*. McGraw-Hill, 1959.